# KOLLAR INJECTIVITY FOR PICARD GROUPS OF PUNCTURED SPECTRA UNDER A DEPTH CONDITION

THE STACKS PROJECT AUTHORS

*Editorial note.* Author statement and proof are retained without a mathematical correction. An earlier assistant raised an unconfirmed concern about the Ext/image wording at source lines 8600–8659; this preparation neither adopts the proposed correction nor adjudicates correctness.

*Editorial note.* Only the original depth/dimension hypotheses are retained. Coherent triples and their polynomial/positivity lemmas are uncounted supporting material.

*Allowed standard prerequisites (editorial).*

- Explicit prerequisite boundary. Local cohomology and normalized dualizing complexes, Matlis duality, depth/support inequalities, Hilbert–Samuel growth in dimension one, existence of a nonvanishing section on finitely many points of a quasi-affine scheme, and catenary S2 dimension theory. Coherent triples and their polynomial/positivity lemmas are retained local proof.

*Source reference.* Original lemma 0BK0: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 14543–14543. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

*Source reference.* Original lemma 01YB: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 2550–2550. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

*Source reference.* Original lemma 00L5: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 775–775. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

*Source reference.* Original lemma 00IV: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 693–693. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

*Source reference.* Original lemma 00IN: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 685–685. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

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*Date*: stacks-project:algebraization.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

*Source reference.* Original lemma 0F20: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 18815–18815. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

*Source reference.* Original lemma 00JZ: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 733–733. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

*Source reference.* Original lemma 090R: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 11419–11419. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

*Source reference.* Original lemma 0AVZ: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 13702–13702. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

*Source reference.* Original lemma 0BLT: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 14606–14606. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

*Source reference.* Original lemma 0EY0: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 18675–18675. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

*Source reference.* Original lemma 00LX: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 803–803. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

*Source reference.* Original lemma 00LK: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 790–790. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

*Source reference.* Original lemma 0BPY: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 14716–14716. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

*Source reference.* Original lemma 0AAK: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 12988–12988. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two

concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

*Source reference.* Original lemma 0DWZ: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 17414–17414. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

*Source reference.* Original lemma 00L8: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 778–778. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

*Source reference.* Original lemma 0A80: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 12898–12898. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

*Source reference.* Original lemma 0FIW: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 19406–19406. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

**Proposition 0.1** (Kollár)**.** [3, Theorem 1.9] *Let $(A, \mathfrak{m})$ be a Noetherian local ring. Let $f \in \mathfrak{m}$. Assume*

(1) *$A$ has a dualizing complex,*
(2) *$f$ is a nonzerodivisor,*
(3) *depth$(A/fA) \geq 2$, or equivalently depth$(A) \geq 3$,*
(4) *if $f \in \mathfrak{p} \subset A$ is a prime ideal with $\dim(A/\mathfrak{p}) = 2$, then depth$(A_\mathfrak{p}) \geq 2$.*

*Let $U$, resp. $U_0$ be the punctured spectrum of $A$, resp. $A/fA$. The map*

$$\mathrm{Pic}(U) \to \mathrm{Pic}(U_0)$$

*is injective. Finally, if (1), (2), (3), $A$ is $(S_2)$, and $\dim(A) \geq 4$, then (4) holds.*

## 1. Coherent triples

Let $(A, \mathfrak{m})$ be a Noetherian local ring. Let $f \in \mathfrak{m}$ be a nonzerodivisor. Set $X = \mathrm{Spec}(A)$, $X_0 = \mathrm{Spec}(A/fA)$, $U = X \setminus V(\mathfrak{m})$, and $U_0 = U \cap X_0$. We say $(\mathcal{F}, \mathcal{F}_0, \alpha)$ is a *coherent triple* if we have

(1) $\mathcal{F}$ is a coherent $\mathcal{O}_U$-module such that $f : \mathcal{F} \to \mathcal{F}$ is injective,
(2) $\mathcal{F}_0$ is a coherent $\mathcal{O}_{X_0}$-module,
(3) $\alpha : \mathcal{F}/f\mathcal{F} \to \mathcal{F}_0|_{U_0}$ is an isomorphism.

There is an obvious notion of a *morphism of coherent triples* which turns the collection of all coherent triples into a category.

The category of coherent triples is additive but not abelian. However, it is clear what a short exact sequence of coherent triples is.

Given two coherent triples $(\mathcal{F}, \mathcal{F}_0, \alpha)$ and $(\mathcal{G}, \mathcal{G}_0, \beta)$ it may not be the case that $(\mathcal{F} \otimes_{\mathcal{O}_U} \mathcal{G}, \mathcal{F}_0 \otimes_{\mathcal{O}_{X_0}} \mathcal{G}_0, \alpha \otimes \beta)$ is a coherent triple[1]. However, if the stalks $\mathcal{G}_x$ are free for all $x \in U_0$, then this does hold.

[1]Namely, it isn't necessarily the case that $f$ is injective on $\mathcal{F} \otimes_{\mathcal{O}_U} \mathcal{G}$.

We will say the coherent triple $(\mathcal{G}, \mathcal{G}_0, \beta)$ is *locally free*, resp. *invertible* if $\mathcal{G}$ and $\mathcal{G}_0$ are locally free, resp. invertible modules. In this case tensoring with $(\mathcal{G}, \mathcal{G}_0, \beta)$ makes sense (see above) and turns short exact sequences of coherent triples into short exact sequences of coherent triples.

**Lemma 1.1.** *For any coherent triple $(\mathcal{F}, \mathcal{F}_0, \alpha)$ there exists a coherent $\mathcal{O}_X$-module $\mathcal{F}'$ such that $f : \mathcal{F}' \to \mathcal{F}'$ is injective, an isomorphism $\alpha' : \mathcal{F}'|_U \to \mathcal{F}$, and a map $\alpha'_0 : \mathcal{F}'/f\mathcal{F}' \to \mathcal{F}_0$ such that $\alpha \circ (\alpha' \bmod f) = \alpha'_0|_{U_0}$.*

*Proof.* Choose a finite $A$-module $M$ such that $\mathcal{F}$ is the restriction to $U$ of the coherent $\mathcal{O}_X$-module associated to $M$, see Local Cohomology, Lemma 0BK0. Since $\mathcal{F}$ is $f$-torsion free, we may replace $M$ by its quotient by $f$-power torsion. On the other hand, let $M_0 = \Gamma(X_0, \mathcal{F}_0)$ so that $\mathcal{F}_0$ is the coherent $\mathcal{O}_{X_0}$-module associated to the finite $A/fA$-module $M_0$. By Cohomology of Schemes, Lemma 01YB there exists an $n$ such that the isomorphism $\alpha_0$ corresponds to an $A/fA$-module homomorphism $\mathfrak{m}^n M/fM \to M_0$ (whose kernel and cokernel are annihilated by a power of $\mathfrak{m}$, but we don't need this). Thus if we take $M' = \mathfrak{m}^n M$ and we let $\mathcal{F}'$ be the coherent $\mathcal{O}_X$-module associated to $M'$, then the lemma is clear. □

Let $(\mathcal{F}, \mathcal{F}_0, \alpha)$ be a coherent triple. Choose $\mathcal{F}', \alpha', \alpha'_0$ as in Lemma 1.1. Set

$$\chi(\mathcal{F}, \mathcal{F}_0, \alpha) = \text{length}_A(\text{Coker}(\alpha'_0)) - \text{length}_A(\text{Ker}(\alpha'_0)) \tag{1.1.1}$$

The expression on the right makes sense as $\alpha'_0$ is an isomorphism over $U_0$ and hence its kernel and coherent are coherent modules supported on $\{\mathfrak{m}\}$ which therefore have finite length (Algebra, Lemma 00L5).

**Lemma 1.2.** *The quantity $\chi(\mathcal{F}, \mathcal{F}_0, \alpha)$ in (1.1.1) does not depend on the choice of $\mathcal{F}', \alpha', \alpha'_0$ as in Lemma 1.1.*

*Proof.* Let $\mathcal{F}', \alpha', \alpha'_0$ and $\mathcal{F}'', \alpha'', \alpha''_0$ be two such choices. For $n > 0$ set $\mathcal{F}'_n = \mathfrak{m}^n \mathcal{F}'$. By Cohomology of Schemes, Lemma 01YB for some $n$ there exists an $\mathcal{O}_X$-module map $\mathcal{F}'_n \to \mathcal{F}''$ agreeing with the identification $\mathcal{F}''|_U = \mathcal{F}'|_U$ determined by $\alpha'$ and $\alpha''$. Then the diagram

$$\begin{array}{ccc} \mathcal{F}'_n/f\mathcal{F}'_n & \longrightarrow & \mathcal{F}'/f\mathcal{F}' \\ \downarrow & & \downarrow{\scriptstyle \alpha'_0} \\ \mathcal{F}''/f\mathcal{F}'' & \xrightarrow{\alpha''_0} & \mathcal{F}_0 \end{array}$$

is commutative after restricting to $U_0$. Hence by Cohomology of Schemes, Lemma 01YB it is commutative after restricting to $\mathfrak{m}^l(\mathcal{F}'_n/f\mathcal{F}'_n)$ for some $l > 0$. Since $\mathcal{F}'_{n+l}/f\mathcal{F}'_{n+l} \to \mathcal{F}'_n/f\mathcal{F}'_n$ factors through $\mathfrak{m}^l(\mathcal{F}'_n/f\mathcal{F}'_n)$ we see that after replacing $n$ by $n + l$ the diagram is commutative. In other words, we have found a third choice $\mathcal{F}''', \alpha''', \alpha'''_0$ such that there are maps $\mathcal{F}''' \to \mathcal{F}''$ and $\mathcal{F}''' \to \mathcal{F}'$ over $X$ compatible with the maps over $U$ and $X_0$. This reduces us to the case discussed in the next paragraph.

Assume we have a map $\mathcal{F}'' \to \mathcal{F}'$ over $X$ compatible with $\alpha', \alpha''$ over $U$ and with $\alpha'_0, \alpha''_0$ over $X_0$. Observe that $\mathcal{F}'' \to \mathcal{F}'$ is injective as it is an isomorphism over $U$ and since $f : \mathcal{F}'' \to \mathcal{F}''$ is injective. Clearly $\mathcal{F}'/\mathcal{F}''$ is supported on $\{\mathfrak{m}\}$ hence has finite length. We have the maps of coherent $\mathcal{O}_{X_0}$-modules

$$\mathcal{F}''/f\mathcal{F}'' \to \mathcal{F}'/f\mathcal{F}' \xrightarrow{\alpha'_0} \mathcal{F}_0$$

whose composition is $\alpha''_0$ and which are isomorphisms over $U_0$. Elementary homological algebra gives a 6-term exact sequence

$$\begin{aligned} 0 \to \text{Ker}(\mathcal{F}''/f\mathcal{F}'' \to \mathcal{F}'/f\mathcal{F}') \to \text{Ker}(\alpha''_0) \to \text{Ker}(\alpha'_0) \to \\ \text{Coker}(\mathcal{F}''/f\mathcal{F}'' \to \mathcal{F}'/f\mathcal{F}') \to \text{Coker}(\alpha''_0) \to \text{Coker}(\alpha'_0) \to 0 \end{aligned}$$

By additivity of lengths (Algebra, Lemma 00IV) we find that it suffices to show that

$$\text{length}_A(\text{Coker}(\mathcal{F}''/f\mathcal{F}'' \to \mathcal{F}'/f\mathcal{F}')) - \text{length}_A(\text{Ker}(\mathcal{F}''/f\mathcal{F}'' \to \mathcal{F}'/f\mathcal{F}')) = 0$$

This follows from applying the snake lemma to the diagram

$$\begin{array}{ccccccccc} 0 & \longrightarrow & \mathcal{F}'' & \xrightarrow{f} & \mathcal{F}'' & \longrightarrow & \mathcal{F}''/f\mathcal{F}'' & \longrightarrow & 0 \\ & & \downarrow & & \downarrow & & \downarrow & & \\ 0 & \longrightarrow & \mathcal{F}' & \xrightarrow{f} & \mathcal{F}' & \longrightarrow & \mathcal{F}'/f\mathcal{F}' & \longrightarrow & 0 \end{array}$$

and the fact that $\mathcal{F}'/\mathcal{F}''$ has finite length. □

**Lemma 1.3.** *We have* $\chi(\mathcal{G}, \mathcal{G}_0, \beta) = \chi(\mathcal{F}, \mathcal{F}_0, \alpha) + \chi(\mathcal{H}, \mathcal{H}_0, \gamma)$ *if*

$$0 \to (\mathcal{F}, \mathcal{F}_0, \alpha) \to (\mathcal{G}, \mathcal{G}_0, \beta) \to (\mathcal{H}, \mathcal{H}_0, \gamma) \to 0$$

*is a short exact sequence of coherent triples.*

*Proof.* Choose $\mathcal{G}', \beta', \beta'_0$ as in Lemma 1.1 for the triple $(\mathcal{G}, \mathcal{G}_0, \beta)$. Denote $j : U \to X$ the inclusion morphism. Let $\mathcal{F}' \subset \mathcal{G}'$ be the kernel of the composition

$$\mathcal{G}' \xrightarrow{\beta'} j_*\mathcal{G} \to j_*\mathcal{H}$$

Observe that $\mathcal{H}' = \mathcal{G}'/\mathcal{F}'$ is a coherent subsheaf of $j_*\mathcal{H}$ and hence $f : \mathcal{H}' \to \mathcal{H}'$ is injective. Hence by the snake lemma we obtain a short exact sequence

$$0 \to \mathcal{F}'/f\mathcal{F}' \to \mathcal{G}'/f\mathcal{G}' \to \mathcal{H}'/f\mathcal{H}' \to 0$$

We have isomorphisms $\alpha' : \mathcal{F}'|_U \to \mathcal{F}$, $\beta' : \mathcal{G}'|_U \to \mathcal{G}$, and $\gamma' : \mathcal{H}'|_U \to \mathcal{H}$ by construction. To finish the proof we'll need to construct maps $\alpha'_0 : \mathcal{F}'/f\mathcal{F}' \to \mathcal{F}_0$ and $\gamma'_0 : \mathcal{H}'/f\mathcal{H}' \to \mathcal{H}_0$ as in Lemma 1.1 and fitting into a commutative diagram

$$\begin{array}{ccccccccc} 0 & \longrightarrow & \mathcal{F}'/f\mathcal{F}' & \longrightarrow & \mathcal{G}'/f\mathcal{G}' & \longrightarrow & \mathcal{H}'/f\mathcal{H}' & \longrightarrow & 0 \\ & & \vdots{\scriptstyle \alpha'_0} & & \downarrow{\scriptstyle \beta'_0} & & \vdots{\scriptstyle \gamma'_0} & & \\ 0 & \longrightarrow & \mathcal{F}_0 & \longrightarrow & \mathcal{G}_0 & \longrightarrow & \mathcal{H}_0 & \longrightarrow & 0 \end{array}$$

However, this may not be possible with our initial choice of $\mathcal{G}'$. From the displayed diagram we see the obstruction is exactly the composition

$$\delta : \mathcal{F}'/f\mathcal{F}' \to \mathcal{G}'/f\mathcal{G}' \xrightarrow{\beta'_0} \mathcal{G}_0 \to \mathcal{H}_0$$

Note that the restriction of $\delta$ to $U_0$ is zero by our choice of $\mathcal{F}'$ and $\mathcal{H}'$. Hence by Cohomology of Schemes, Lemma 01YB there exists an $k > 0$ such that $\delta$ vanishes on $\mathfrak{m}^k \cdot (\mathcal{F}'/f\mathcal{F}')$. For $n > k$ set $\mathcal{G}'_n = \mathfrak{m}^n\mathcal{G}'$, $\mathcal{F}'_n = \mathcal{G}'_n \cap \mathcal{F}'$, and $\mathcal{H}'_n = \mathcal{G}'_n/\mathcal{F}'_n$. Observe that $\beta'_0$ can be composed with $\mathcal{G}'_n/f\mathcal{G}'_n \to \mathcal{G}'/f\mathcal{G}'$ to give a map $\beta'_{n,0} : \mathcal{G}'_n/f\mathcal{G}'_n \to \mathcal{G}_0$ as in Lemma 1.1. By Artin-Rees (Algebra, Lemma 00IN) we may choose $n$ such that $\mathcal{F}'_n \subset \mathfrak{m}^k\mathcal{F}'$. As above the maps $f : \mathcal{F}'_n \to \mathcal{F}'_n$, $f : \mathcal{G}'_n \to \mathcal{G}'_n$, and $f : \mathcal{H}'_n \to \mathcal{H}'_n$ are injective and as above using the snake lemma we obtain a short exact sequence

$$0 \to \mathcal{F}'_n/f\mathcal{F}'_n \to \mathcal{G}'_n/f\mathcal{G}'_n \to \mathcal{H}'_n/f\mathcal{H}'_n \to 0$$

As above we have isomorphisms $\alpha'_n : \mathcal{F}'_n|_U \to \mathcal{F}$, $\beta'_n : \mathcal{G}'_n|_U \to \mathcal{G}$, and $\gamma'_n : \mathcal{H}'_n|_U \to \mathcal{H}$. We consider the obstruction

$$\delta_n : \mathcal{F}'_n/f\mathcal{F}'_n \to \mathcal{G}'_n/f\mathcal{G}'_n \xrightarrow{\beta'_{n,0}} \mathcal{G}_0 \to \mathcal{H}_0$$

as before. However, the commutative diagram

$$\begin{array}{ccccccc} \mathcal{F}'_n/f\mathcal{F}'_n & \longrightarrow & \mathcal{G}'_n/f\mathcal{G}'_n & \xrightarrow[\beta'_{n,0}]{} & \mathcal{G}_0 & \longrightarrow & \mathcal{H}_0 \\ \downarrow & & \downarrow & & \downarrow & & \downarrow \\ \mathcal{F}'/f\mathcal{F}' & \longrightarrow & \mathcal{G}'/f\mathcal{G}' & \xrightarrow{\beta'_0} & \mathcal{G}_0 & \longrightarrow & \mathcal{H}_0 \end{array}$$

our choice of $n$ and our observation about $\delta$ show that $\delta_n = 0$. This produces the desired maps $\alpha'_{n,0} : \mathcal{F}'_n/f\mathcal{F}'_n \to \mathcal{F}_0$, and $\gamma'_{n,0} : \mathcal{H}'_n/f\mathcal{H}'_n \to \mathcal{H}_0$. OK, so we may use $\mathcal{F}'_n, \alpha'_n, \alpha'_{n,0}$, $\mathcal{G}'_n, \beta'_n, \beta'_{n,0}$, and $\mathcal{H}'_n, \gamma'_n, \gamma'_{n,0}$ to compute $\chi(\mathcal{F}, \mathcal{F}_0, \alpha)$, $\chi(\mathcal{G}, \mathcal{G}_0, \beta)$, and $\chi(\mathcal{H}, \mathcal{H}_0, \gamma)$. Now finally the lemma follows from an application of the snake lemma to

$$\begin{array}{ccccccccc} 0 & \longrightarrow & \mathcal{F}'_n/f\mathcal{F}'_n & \longrightarrow & \mathcal{G}'_n/f\mathcal{G}'_n & \longrightarrow & \mathcal{H}'_n/f\mathcal{H}'_n & \longrightarrow & 0 \\ & & \downarrow & & \downarrow & & \downarrow & & \\ 0 & \longrightarrow & \mathcal{F}_0 & \longrightarrow & \mathcal{G}_0 & \longrightarrow & \mathcal{H}_0 & \longrightarrow & 0 \end{array}$$

and additivity of lengths (Algebra, Lemma 00IV). □

**Proposition 1.4.** *Let $(\mathcal{F}, \mathcal{F}_0, \alpha)$ be a coherent triple. Let $(\mathcal{L}, \mathcal{L}_0, \lambda)$ be an invertible coherent triple. Then the function*

$$\mathbf{Z} \longrightarrow \mathbf{Z}, \quad n \longmapsto \chi((\mathcal{F}, \mathcal{F}_0, \alpha) \otimes (\mathcal{L}, \mathcal{L}_0, \lambda)^{\otimes n})$$

*is a polynomial of degree $\leq \dim(\mathit{Supp}(\mathcal{F}))$.*

More precisely, if $\mathcal{F} = 0$, then the function is constant. If $\mathcal{F}$ has finite support in $U$, then the function is constant. If the support of $\mathcal{F}$ in $U$ has dimension 1, i.e., the closure of the support of $\mathcal{F}$ in $X$ has dimension 2, then the function is linear, etc.

*Proof.* We will prove this by induction on the dimension of the support of $\mathcal{F}$.

The base case is when $\mathcal{F} = 0$. Then either $\mathcal{F}_0$ is zero or its support is $\{\mathfrak{m}\}$. In this case we have

$$(\mathcal{F}, \mathcal{F}_0, \alpha) \otimes (\mathcal{L}, \mathcal{L}_0, \lambda)^{\otimes n} = (0, \mathcal{F}_0 \otimes \mathcal{L}_0^{\otimes n}, 0) \cong (0, \mathcal{F}_0, 0)$$

Thus the function of the lemma is constant with value equal to the length of $\mathcal{F}_0$.

Induction step. Assume the support of $\mathcal{F}$ is nonempty. Let $\mathcal{G}_0 \subset \mathcal{F}_0$ denote the submodule of sections supported on $\{\mathfrak{m}\}$. Then we get a short exact sequence

$$0 \to (0, \mathcal{G}_0, 0) \to (\mathcal{F}, \mathcal{F}_0, \alpha) \to (\mathcal{F}, \mathcal{F}_0/\mathcal{G}_0, \alpha) \to 0$$

This sequence remains exact if we tensor by the invertible coherent triple $(\mathcal{L}, \mathcal{L}_0, \lambda)$, see discussion above. Thus by additivity of $\chi$ (Lemma 1.3) and the base case explained above, it suffices to prove the induction step for $(\mathcal{F}, \mathcal{F}_0/\mathcal{G}_0, \alpha)$. In this way we see that we may assume $\mathfrak{m}$ is not an associated point of $\mathcal{F}_0$.

Let $T = \mathrm{Ass}(\mathcal{F}) \cup \mathrm{Ass}(\mathcal{F}/f\mathcal{F})$. Since $U$ is quasi-affine, we can find $s \in \Gamma(U, \mathcal{L})$ which does not vanish at any $u \in T$, see Properties, Lemma 0F20. After multiplying $s$ by a suitable element of $\mathfrak{m}$ we may assume $\lambda(s \bmod f) = s_0|_{U_0}$ for some $s_0 \in \Gamma(X_0, \mathcal{L}_0)$; details omitted. We obtain a morphism

$$(s, s_0) : (\mathcal{O}_U, \mathcal{O}_{X_0}, 1) \longrightarrow (\mathcal{L}, \mathcal{L}_0, \lambda)$$

in the category of coherent triples. Let $\mathcal{G} = \mathrm{Coker}(s : \mathcal{F} \to \mathcal{F} \otimes \mathcal{L})$ and $\mathcal{G}_0 = \mathrm{Coker}(s_0 : \mathcal{F}_0 \to \mathcal{F}_0 \otimes \mathcal{L}_0)$. Observe that $s_0 : \mathcal{F}_0 \to \mathcal{F}_0 \otimes \mathcal{L}_0$ is injective as it is injective on $U_0$ by our choice of $s$ and as $\mathfrak{m}$ isn't an associated point of $\mathcal{F}_0$. It

follows that there exists an isomorphism $\beta : \mathcal{G}/f\mathcal{G} \to \mathcal{G}_0|_{U_0}$ such that we obtain a short exact sequence

$$0 \to (\mathcal{F}, \mathcal{F}_0, \alpha) \to (\mathcal{F}, \mathcal{F}_0, \alpha) \otimes (\mathcal{L}, \mathcal{L}_0, \lambda) \to (\mathcal{G}, \mathcal{G}_0, \beta) \to 0$$

By induction on the dimension of the support we know the proposition holds for the coherent triple $(\mathcal{G}, \mathcal{G}_0, \beta)$. Using the additivity of Lemma 1.3 we see that

$$n \longmapsto \chi((\mathcal{F}, \mathcal{F}_0, \alpha) \otimes (\mathcal{L}, \mathcal{L}_0, \lambda)^{\otimes n+1}) - \chi((\mathcal{F}, \mathcal{F}_0, \alpha) \otimes (\mathcal{L}, \mathcal{L}_0, \lambda)^{\otimes n})$$

is a polynomial. We conclude by a variant of Algebra, Lemma 00JZ for functions defined for all integers (details omitted). □

**Lemma 1.5.** *Assume depth$(A) \geq 3$ or equivalently depth$(A/fA) \geq 2$. Let $(\mathcal{L}, \mathcal{L}_0, \lambda)$ be an invertible coherent triple. Then*

$$\chi(\mathcal{L}, \mathcal{L}_0, \lambda) = \mathit{length}_A \operatorname{Coker}(\Gamma(U, \mathcal{L}) \to \Gamma(U_0, \mathcal{L}_0))$$

*and in particular this is $\geq 0$. Moreover, $\chi(\mathcal{L}, \mathcal{L}_0, \lambda) = 0$ if and only if $\mathcal{L} \cong \mathcal{O}_U$.*

*Proof.* The equivalence of the depth conditions follows from Algebra, Lemma 090R. By the depth condition we see that $\Gamma(U, \mathcal{O}_U) = A$ and $\Gamma(U_0, \mathcal{O}_{U_0}) = A/fA$, see Dualizing Complexes, Lemma 0AVZ and Local Cohomology, Lemma 0BK0. Using Local Cohomology, Lemma 0BLT we find that $M = \Gamma(U, \mathcal{L})$ is a finite $A$-module. This in turn implies $\operatorname{depth}(M) \geq 2$ for example by part (4) of Local Cohomology, Lemma 0BK0 or by Divisors, Lemma 0EY0. Also, we have $\mathcal{L}_0 \cong \mathcal{O}_{X_0}$ as $X_0$ is a local scheme. Hence we also see that $M_0 = \Gamma(X_0, \mathcal{L}_0) = \Gamma(U_0, \mathcal{L}_0|_{U_0})$ and that this module is isomorphic to $A/fA$.

By the above $\mathcal{F}' = \widetilde{M}$ is a coherent $\mathcal{O}_X$-module whose restriction to $U$ is isomorphic to $\mathcal{L}$. The isomorphism $\lambda : \mathcal{L}/f\mathcal{L} \to \mathcal{L}_0|_{U_0}$ determines a map $M/fM \to M_0$ on global sections which is an isomorphism over $U_0$. Since $\operatorname{depth}(M) \geq 2$ we see that $H^0_{\mathfrak{m}}(M/fM) = 0$ and it follows that $M/fM \to M_0$ is injective. Thus by definition

$$\chi(\mathcal{L}, \mathcal{L}_0, \lambda) = \operatorname{length}_A \operatorname{Coker}(M/fM \to M_0)$$

which gives the first statement of the lemma.

Finally, if this length is 0, then $M \to M_0$ is surjective. Hence we can find $s \in M = \Gamma(U, \mathcal{L})$ mapping to a trivializing section of $\mathcal{L}_0$. Consider the finite $A$-modules $K$, $Q$ defined by the exact sequence

$$0 \to K \to A \xrightarrow{s} M \to Q \to 0$$

The supports of $K$ and $Q$ do not meet $U_0$ because $s$ is nonzero at points of $U_0$. Using Algebra, Lemma 00LX we see that $\operatorname{depth}(K) \geq 2$ (observe that $As \subset M$ has depth $\geq 1$ as a submodule of $M$). Thus the support of $K$ if nonempty has dimension $\geq 2$ by Algebra, Lemma 00LK. This contradicts $\operatorname{Supp}(M) \cap V(f) \subset \{\mathfrak{m}\}$ unless $K = 0$. When $K = 0$ we find that $\operatorname{depth}(Q) \geq 2$ and we conclude $Q = 0$ as before. Hence $A \cong M$ and $\mathcal{L}$ is trivial. □

## 2. Invertible modules on punctured spectra, I

In this section we prove some local Lefschetz theorems for the Picard group. Some of the ideas are taken from [2], [1], and [3].

**Lemma 2.1.** *Let $(A, \mathfrak{m})$ be a Noetherian local ring. Let $f \in \mathfrak{m}$ be a nonzerodivisor and assume that depth$(A/fA) \geq 2$, or equivalently depth$(A) \geq 3$. Let $U$, resp. $U_0$ be the punctured spectrum of $A$, resp. $A/fA$. The map*

$$\operatorname{Pic}(U) \to \operatorname{Pic}(U_0)$$

*is injective on torsion.*

*Proof.* Let $\mathcal{L}$ be an invertible $\mathcal{O}_U$-module. Observe that $\mathcal{L}$ maps to 0 in $\mathrm{Pic}(U_0)$ if and only if we can extend $\mathcal{L}$ to an invertible coherent triple $(\mathcal{L}, \mathcal{L}_0, \lambda)$ as in Section 1. By Proposition 1.4 the function

$$n \longmapsto \chi((\mathcal{L}, \mathcal{L}_0, \lambda)^{\otimes n})$$

is a polynomial. By Lemma 1.5 the value of this polynomial is zero if and only if $\mathcal{L}^{\otimes n}$ is trivial. Thus if $\mathcal{L}$ is torsion, then this polynomial has infinitely many zeros, hence is identically zero, hence $\mathcal{L}$ is trivial. □

*Proof.* Let $\mathcal{L}$ be an invertible $\mathcal{O}_U$-module. Observe that $\mathcal{L}$ maps to 0 in $\mathrm{Pic}(U_0)$ if and only if we can extend $\mathcal{L}$ to an invertible coherent triple $(\mathcal{L}, \mathcal{L}_0, \lambda)$ as in Section 1. By Proposition 1.4 the function

$$n \longmapsto \chi((\mathcal{L}, \mathcal{L}_0, \lambda)^{\otimes n})$$

is a polynomial $P$. By Lemma 1.5 we have $P(n) \geq 0$ for all $n \in \mathbf{Z}$ with equality if and only if $\mathcal{L}^{\otimes n}$ is trivial. In particular $P(0) = 0$ and $P$ is either identically zero and we win or $P$ has even degree $\geq 2$.

Set $M = \Gamma(U, \mathcal{L})$ and $M_0 = \Gamma(X_0, \mathcal{L}_0) = \Gamma(U_0, \mathcal{L}_0)$. Then $M$ is a finite $A$-module of depth $\geq 2$ and $M_0 \cong A/fA$, see proof of Lemma 1.5. Note that $H^2_{\mathfrak{m}}(M)$ is finite $A$-module by Local Cohomology, Lemma 0BPY and the fact that $H^i_{\mathfrak{m}}(A) = 0$ for $i = 0, 1, 2$ since $\mathrm{depth}(A) \geq 3$. Consider the short exact sequence

$$0 \to M/fM \to M_0 \to Q \to 0$$

Lemma 1.5 tells us $Q$ has finite length equal to $\chi(\mathcal{L}, \mathcal{L}_0, \lambda)$. We obtain $Q = H^1_{\mathfrak{m}}(M/fM)$ and $H^i_{\mathfrak{m}}(M/fM) = H^i_{\mathfrak{m}}(M_0) \cong H^i_{\mathfrak{m}}(A/fA)$ for $i > 1$ from the long exact sequence of local cohomology associated to the displayed short exact sequence. Consider the long exact sequence of local cohomology associated to the sequence $0 \to M \to M \to M/fM \to 0$. It starts with

$$0 \to Q \to H^2_{\mathfrak{m}}(M) \to H^2_{\mathfrak{m}}(M) \to H^2_{\mathfrak{m}}(A/fA)$$

Using additivity of lengths we see that $\chi(\mathcal{L}, \mathcal{L}_0, \lambda)$ is equal to the length of the image of $H^2_{\mathfrak{m}}(M) \to H^2_{\mathfrak{m}}(A/fA)$.

Let prove the lemma in a special case to elucidate the rest of the proof. Namely, assume for a moment that $H^2_{\mathfrak{m}}(A/fA)$ is a finite length module. Then we would have $P(1) \leq \mathrm{length}_A H^2_{\mathfrak{m}}(A/fA)$. The exact same argument applied to $\mathcal{L}^{\otimes n}$ shows that $P(n) \leq \mathrm{length}_A H^2_{\mathfrak{m}}(A/fA)$ for all $n$. Thus $P$ cannot have positive degree and we win. In the rest of the proof we will modify this argument to give a linear upper bound for $P(n)$ which suffices.

Let us study the map $H^2_{\mathfrak{m}}(M) \to H^2_{\mathfrak{m}}(M_0) \cong H^2_{\mathfrak{m}}(A/fA)$. Choose a normalized dualizing complex $\omega_A^\bullet$ for $A$. By local duality (Dualizing Complexes, Lemma 0AAK) this map is Matlis dual to the map

$$\mathrm{Ext}^{-2}_A(M, \omega_A^\bullet) \longleftarrow \mathrm{Ext}^{-2}_A(M_0, \omega_A^\bullet)$$

whose image therefore has the same (finite) length. The support (if nonempty) of the finite $A$-module $\mathrm{Ext}^{-2}_A(M_0, \omega_A^\bullet)$ consists of $\mathfrak{m}$ and a finite number of primes $\mathfrak{p}_1, \ldots, \mathfrak{p}_r$ containing $f$ with $\dim(A/\mathfrak{p}_i) = 1$. Namely, by Local Cohomology, Lemma 0DWZ the support is contained in the set of primes $\mathfrak{p} \subset A$ with $\mathrm{depth}_{A_\mathfrak{p}}(M_{0,\mathfrak{p}}) + \dim(A/\mathfrak{p}) \leq 2$. Thus it suffices to show there is no prime $\mathfrak{p}$ containing $f$ with $\dim(A/\mathfrak{p}) = 2$ and $\mathrm{depth}_{A_\mathfrak{p}}(M_{0,\mathfrak{p}}) = 0$. However, because $M_{0,\mathfrak{p}} \cong (A/fA)_\mathfrak{p}$ this would give $\mathrm{depth}(A_\mathfrak{p}) = 1$ which contradicts assumption (4). Choose a section $t \in \Gamma(U, \mathcal{L}^{\otimes -1})$ which does not vanish in the points $\mathfrak{p}_1, \ldots, \mathfrak{p}_r$, see Properties, Lemma 0F20. Multiplication by $t$ on global sections determines a map $t : M \to A$ which defines an isomorphism $M_{\mathfrak{p}_i} \to A_{\mathfrak{p}_i}$ for $i = 1, \ldots, r$. Denote $t_0 = t|_{U_0}$ the

corresponding section of $\Gamma(U_0, \mathcal{L}_0^{\otimes -1})$ which similarly determines a map $t_0 : M_0 \to A/fA$ compatible with $t$. We conclude that there is a commutative diagram

$$\begin{array}{ccc} \mathrm{Ext}_A^{-2}(M, \omega_A^\bullet) & \longleftarrow & \mathrm{Ext}_A^{-2}(M_0, \omega_A^\bullet) \\ \uparrow{\scriptstyle t} & & \uparrow{\scriptstyle t_0} \\ \mathrm{Ext}_A^{-2}(A, \omega_A^\bullet) & \longleftarrow & \mathrm{Ext}_A^{-2}(A/fA, \omega_A^\bullet) \end{array}$$

It follows that the length of the image of the top horizontal map is at most the length of $\mathrm{Ext}_A^{-2}(A/fA, \omega_A^\bullet)$ plus the length of the cokernel of $t_0$.

However, if we replace $\mathcal{L}$ by $\mathcal{L}^n$ for $n > 1$, then we can use

$$t^n : M_n = \Gamma(U, \mathcal{L}^{\otimes n}) \longrightarrow \Gamma(U, \mathcal{O}_U) = A$$

instead of $t$. This replaces $t_0$ by its $n$th power. Thus the length of the image of the map $\mathrm{Ext}_A^{-2}(M_n, \omega_A^\bullet) \leftarrow \mathrm{Ext}_A^{-2}(M_{n,0}, \omega_A^\bullet)$ is at most the length of $\mathrm{Ext}_A^{-2}(A/fA, \omega_A^\bullet)$ plus the length of the cokernel of

$$t_0^n : \mathrm{Ext}_A^{-2}(A/fA, \omega_A^\bullet) \longrightarrow \mathrm{Ext}_A^{-2}(M_{n,0}, \omega_A^\bullet)$$

Via the isomorphism $M_0 \cong A/fA$ the map $t_0$ becomes $g : A/fA \to A/fA$ for some $g \in A/fA$ and via the corresponding isomorphisms $M_{n,0} \cong A/fA$ the map $t_0^n$ becomes $g^n : A/fA \to A/fA$. Thus the length of the cokernel above is the length of the quotient of $\mathrm{Ext}_A^{-2}(A/fA, \omega_A^\bullet)$ by $g^n$. Since $\mathrm{Ext}_A^{-2}(A/fA, \omega_A^\bullet)$ is a finite $A$-module with support $T$ of dimension 1 and since $V(g) \cap T$ consists of the closed point by our choice of $t$ this length grows linearly in $n$ by Algebra, Lemma 00L8.

To finish the proof we prove the final assertion. Assume $f \in \mathfrak{m} \subset A$ satisfies (1), (2), (3), $A$ is $(S_2)$, and $\dim(A) \geq 4$. Condition (1) implies $A$ is catenary, see Dualizing Complexes, Lemma 0A80. Then $\mathrm{Spec}(A)$ is equidimensional by Local Cohomology, Lemma 0FIW. Thus $\dim(A_\mathfrak{p}) + \dim(A/\mathfrak{p}) \geq 4$ for every prime $\mathfrak{p}$ of $A$. Then $\mathrm{depth}(A_\mathfrak{p}) \geq \min(2, \dim(A_\mathfrak{p})) \geq \min(2, 4 - \dim(A/\mathfrak{p}))$ and hence (4) holds. □

## References


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